# The Chelsea Eats Program: Experimental Impacts

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## Abstract

In the midst of the 2020 Covid economic shutdown, 2213 households from a highly food-insecure population in Chelsea, Massachusetts were selected by lottery to receive a cash card that was restocked with up to $400 each month for nine months. This program, called Chelsea Eats, provides an opportunity to assess the causal effects of income on food consumption, financial well-being, and a variety of potential downstream impacts using a randomized controlled design. As 15 percent of Chelsea households received a cash card, it also offers the opportunity to assess some of the broader impacts of a basic income program operating at scale. We find that the treatment group had higher food expenditures, greater consumption of fish and fresh meat, and greater food satisfaction than the control group. The treatment group also had lower levels of financial distress. In early months of the program, we observed a decline in food insecurity for the treatment group compared with the control group, but that difference did not persist. There were no statistically significant treatment-control differences in the two pre-specified primary downstream outcomes: self-reported physical/mental health and child school attendance. Receiving the cash cards did not reduce employment or hours of work. Among secondary outcomes (results that should be seen as tentative because of the risk of type I error when many endpoints are examined), there was a reduction in residential moves and an increase in pregnancy.

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## 1. Introduction

Chelsea, Massachusetts, a city of 40,000 people just north of Boston, was among the places in the U.S. hit hardest by the initial wave of COVID-19. By April 2020, 32 percent of asymptomatic Chelsea residents who were tested at a mobile site had SARS-CoV-2 antibodies, the highest seroprevalence observed in the U.S. as of that date (Naranbhai et al, 2020). That same month, as the economy shut down, Chelsea's unemployment rate reached 21.2 percent, up from 3.4 percent in February.[1]

Chelsea is one of the lowest-income municipalities in Massachusetts, has the second highest population density, and the largest foreign-born population share.[2] Chelsea's residents include many undocumented immigrants, mostly from Latin America, who were ineligible for unemployment insurance, stimulus checks, or SNAP. In response to the health and economic crisis caused by the pandemic, the City of Chelsea began a large food relief effort, giving away 800 to 900 boxes of food per day. After five months of running food distribution sites, the City decided to redirect its efforts toward distributing financial support so that residents could purchase food and other essentials on their own. Through a program called "Chelsea Eats," 2213 applicants were selected by lottery to receive cash cards from among 3615 applicants. Approximately 15 percent of Chelsea households received Chelsea Eats cards.[3] The first payments occurred in late November 2020, and cards were restocked each month for a total of

[1] U.S. Bureau of Labor Statistics, Local Area Unemployment Statistics, series LAUCT251320500000006, extracted November 19, 2022.
[2] American Community Survey; Massachusetts Immigrant and Refugee Advocacy Coalition (2019)..
[3] Dividing the 2213 recipient households by the official U.S. Census estimate that Chelsea had 13,174 households in 2020, would suggest that nearly 17 percent of Chelsea households received Chelsea Eats cards. In coming up with our 15 percent estimate, we multiply the Census-based estimate by 40,615/45,000, to adjust for likely undercounting of undocumented residents (40,615 is the Census population estimate for 2020 and 45,000 is a typical guess used in the community of how many people actually live in Chelsea).

nine months. Most households received $400 per month, but one- and two-person households received $200 and $300 respectively. The cards could be spent anywhere that Visa was accepted.

The lottery makes it possible to evaluate the impact of the Chelsea Eats program on food consumption, financial well-being, and a variety of potential downstream impacts using a randomized controlled design. The level of economic distress experienced in our Chelsea sample is very unusual in a U.S. context. For example, at baseline 51 percent of program applicants reported that their family "sometimes" or "often" did not have enough food to eat, seven times the Massachusetts-wide rate in the same month as measured by the U.S. Census Pulse Survey. Understanding the impact of this relief program is, therefore, inherently interesting. But beyond an assessment of this particular program in this particular context, the experiment offers insights into four important economic questions:

- First, what are the tradeoffs in providing food assistance in cash rather than in kind? A long tradition in economics, with prominent examples such as Milton Friedman's advocacy for the negative income tax and Joel Waldfogel's calculations of the deadweight loss of gift giving, focuses on the welfare gains from letting people make their own consumption choices.[4] Yet important parts of the U.S. food assistance system provide food directly (FEMA disaster assistance, food pantries, free school lunches) or with restrictions on choices (SNAP, WIC), and a large literature has studied the impact of these design choices on food consumption and other outcomes.[5] The Chelsea Eats program gives us the opportunity to study the benefits of targeting food insecurity on a community-wide level using cash assistance in a context in which other forms of relief

[4] Friedman (1962); Waldfogel (1993).
[5] Prominent U.S.-focused examples include (Fraker 1995) and Hoynes and Schanzenbach (2009). See Fraker (1990) for a survey. For a global perspective see Alderman, Gentilini, and Yemtsov (2018).

such as food pantries continue to exist, and to compare the resource costs of cash assistance to an in-kind approach.

- Second, what is the value of the U.S. safety net? It is quite unusual to be able to study the causal impact of providing money at a time of economic need to a large sample of U.S. residents who, because of their immigration status, lack access to the main U.S. safety net programs. Most of what we know about the impact of the U.S. safety net comes from looking at incremental impacts of altering benefit levels (e.g., Gruber, 1997; Schmidt, Shore-Sheppard, and Watson, 2016), examining what happens when time-limited benefits expire (e.g., Ganong and Noel, 2019), or studying the introduction of programs (e.g., Hoynes, Schanzenbach, and Almond, 2016). While the incremental studies are valuable – since most policy debates are about changes in benefit levels rather than eliminating programs altogether, and the more historical studies can teach us about long-run effects – it is informative to have a modern assessment of the impact of assistance starting from a much lower base. Moreover, the results in this paper are directly relevant to questions about the value of extending safety net programs to undocumented workers, and about immigration reforms that would offer undocumented workers a path to citizenship.
- Third, what is the relationship between income and diet? A large literature documents that lower-income populations in the U.S. typically have less healthy diets (e.g., Mancino, Guthrie, Ver Ploeg, and Lin, 2018). While the literature posits several plausible channels through which insufficient income might lead to poor diet, including affordability, access to grocery stores, and stress (Darmon and Drewnowski, 2008; Laraia et al, 2017), other research suggests that education and culture may be more important

factors than the ability to afford healthier food in determining diet quality (Alhoff et al, 2022; Akresh, 2007). The limited number of studies that go beyond establishing correlations to examine causal relationships have often found that income plays a minor role (e.g, Gregory et al, 2013; Hastings and Shapiro, 2021). Thus, the ability to assess the impact of income on diet for a low-income population in a randomized controlled setting is valuable.

- Fourth, what is the impact of providing a basic income community-wide? There has been a resurgence in interest in basic income programs in the past several years, but most of the evaluation evidence to date comes from very small pilots where the minimum detectable effects are quite large. [6] For example, in the Stockton, California study, the treatment group contained only 125 individuals (West, Castro Baker, Samra, and Coltrera, 2021). While several larger pilots have recently launched, these are mostly in large cities or draw sample members from across the country, meaning that none to our knowledge will match the Chelsea program in providing benefits to 15 percent of households in one community. Thus, Chelsea Eats provides the best evidence we are likely to have for some time on what a basic income program looks like at scale in the U.S. – for example in the extent to which it replaces informal safety nets within a community.[7]

Our main data for this study come from six surveys administered to lottery entrants between August 2020 and May 2021. Prior to randomization, 1936 sample members enrolled in

[6] There have also been some larger recent studies of one-time cash payments. Jacob et al (2022) find no effects of $1000 payments to low-income households on a variety of outcomes. Jaroszewicz et al (2022) find no effects of $500 and $2000 payments.

[7] See Hoynes and Rothstein (2018) on what questions about guaranteed income programs can be answered in small short-term pilots. Jones and Marinescu's (2022) study of the Alaska Permanent Fund measures labor market impacts of a fully scaled permanent cash transfer program.

our study population. We were highly successful in avoiding attrition, with 95 percent of our study population completing our final survey in April/May 2021. We also analyze administrative records on school attendance, information on the stores and vendors where lottery winners spent their cash cards, and information on where both treatment and control group members spent the smaller $20 gift cards we provided sample members for completing each survey round.

To maintain statistical power in the presence of a large number of measured outcomes, we pre-specified as our primary outcomes five indices that summarize measures within domains (Liebman, 2021). Three of these indices (food satisfaction, food security, and financial distress) measured the direct effects of the intervention, while the remaining two (health and school attendance) focused on potential downstream outcomes.

We find that the treatment group had higher food expenditures, greater consumption of fish and fresh meat, and greater food satisfaction than the control group. The treatment group also had lower levels of financial distress. In early months of the program we observed a decline in food insecurity for the treatment group compared with the control group, but that difference did not persist. There were no statistically significant treatment-control differences in the two pre-specified primary downstream outcomes: self-reported physical/mental health and child school attendance. Receiving the cash cards did not reduce employment or hours of work. Among secondary outcomes (results that should be seen as tentative because of the risk of type I error when many endpoints are examined), there was a reduction in residential moves and an increase in pregnancy.

The rest of the paper is structured as follows. Section two provides additional background on the Chelsea context and the lottery implementation. Section three describes our

data and analytic methods. Section four presents descriptive results on baseline characteristics, the evolution of control group outcomes over the course of the study, and Chelsea Eats card spending. Section five presents results for primary outcomes. Section six presents results for secondary outcomes. Section seven concludes.

## 2. Chelsea context and lottery implementation

The City of Chelsea was hit extremely hard from both a health and economic perspective by the initial wave of COVID-19 that occurred after the outbreak from the February 2020 Biogen conference at the Boston Marriott Long Wharf hotel.[8] The City's high population density, with many families living doubled and tripled up in apartments, and large number of residents who were employed in front line service sector jobs or who commuted via public transportation, led to high infection risk. Through December 2020, Chelsea had the highest rate of confirmed COVID-19 cases in Massachusetts – 11.9 per 100.[9] In addition, the City was particularly vulnerable to the COVID economic shock both because its population was disproportionately employed in sectors of the economy that completely shut down (such as restaurants and office cleaning), and because its large number of suddenly unemployed undocumented residents were not eligible for most types of federal relief.

Local community organizations and the City of Chelsea responded to the economic crisis by mounting a large food distribution effort. During the second week of April 2020, the City of Chelsea began running food distribution sites. The City established a centralized logistics hub where food products were procured, stored, and boxed for distribution. The City ran two neighborhood pop-up pantries five days each week.[10] These sites rotated daily to different

[8] Lemieux et al (2022).
[9] Commonwealth of Massachusetts (2020).
[10] During the first month of operation, these sites were run in partnership with the National Guard.

locations in Chelsea. The City also operated a home delivery service for households afflicted by COVID-19 as well as for other vulnerable populations. At the height of its operation, the City produced and distributed 800 to 900 boxes of food per day, enough to feed 10,000 residents on a weekly basis.

Running a city-wide food distribution program at this scale was putting a lot of strain on Chelsea's municipal resources, so the City decided to redirect its efforts toward distributing financial support to individuals so that residents could purchase their own food. They called this program Chelsea Eats.

The City closed down its pop-up food pantries on September 14th, 2020 after five months of operations. However, community organizations continued to distribute more than 10,000 boxes of free groceries and more than 30,000 free prepared meals each week. Demand for free food remained high. As of January 2021, it was still common to see queues forming that were more than 600 feet in length.

The City announced the Chelsea Eats program on July 29, 2020 and invited residents to enroll in the lottery to receive the cash cards. Initial funding was approximately $4 million and came from the City's general fund, Commonwealth of Massachusetts Cares Act funds, and philanthropic contributions.[11] City workers with tablet computers walked along the lines at food distribution sites, signing residents up for the lottery. Community groups also assisted residents in applying, and residents could apply directly on the City website or by submitting a paper application. Applications closed on August 17, 2020. The lottery occurred at 5:25pm on September 17, 2020 and was broadcast live on the City's Facebook page.

[11] Ultimately, another $2 million was raised, allowing the program to run for 9 months rather than 6 months.

The City announced that, "Families with children, disabled residents, veterans, seniors over 65, households that have been economically impacted by COVID-19, and households that are ineligible for other forms of federal assistance, such as SNAP and WIC, will have preference" in the lottery. In practice, these preferences were implemented by giving families a different number of lottery "tickets" depending on how many criteria they met. Every entrant received an initial ticket and then entrants received additional tickets for meeting each of up to eight additional criteria. These criteria can be seen in rows three to ten of Appendix Table A.1, which contains characteristics of lottery applicants from their application forms. Applicants received additional lottery tickets if they had at least one child age 0 to 5 in their family, had at least one child age 6 to 17 in their family, had at least one family member age 65+, and so on.

The first three columns of Appendix Table A.1 show the characteristics of all lottery applicants and of winners and losers. The first two rows of column 1 show that mean family size for the entire sample was 3.2 and mean family income was $17,040. These two measures determined the amount of cash families were eligible for and whether they were income-eligible for the program but did not affect the number of lottery tickets they received. The next eight rows show the eight criteria that determined the number of lottery tickets a family received. The second to last row of column 1 shows that families received 4.3 lottery tickets on average.

Comparing columns 2 and 3, lottery winners are more likely than lottery losers to exhibit the characteristics that resulted in receiving additional lottery tickets, as would be expected given the weighted nature of the lottery. Column 4 shows the characteristics of households that were unsuccessful in winning the lottery, weighted by their inverse probability of winning the lottery so as to match the distribution of the lottery winners by sample characteristics.[12] The t-test P-

[12] The control group weights that balance the sample are (probability of winning the lottery/(1-probability of winning the lottery)).

values in column 5 show no statistically significant differences between lottery winners and lottery losers, consistent with successful randomization.

Chelsea Eats cards were credited for the first time on November 25, 2020 and then credited once per month, with the final payments made in early August 2021.[13] Initially, the program was intended to run for six months, but as the economic situation in Chelsea continued to be difficult into 2021, the City and its philanthropic partners decided to extend the program for an extra three months.

At the time of the lottery, the City also created a randomized waitlist that was not publicized but could be used to allocate unused cards in case any lottery winners were determined to be ineligible or did not claim their cards. Ultimately, 139 applicants from the waitlist were offered Chelsea Eats cards on top of the initial 2074 lottery winners. Thus, there were 2213 total lottery winners and 1402 lottery losers. In the analysis that follows, all lottery winners remain in the treatment group, even if they were disqualified, since there is no way to remove the equivalent individuals from the control group. Individuals who received cards from the waitlist are also considered members of the treatment group. While waitlist individuals did not receive their cards until December, their cards were credited with both the November and December payments at that time.[14]

[13] The dates that cards were credited appear to have been 11/25/20, 12/22/20, 1/29/21, 2/26/21, 4/2/21, 4/30/21, 6/1/21, 7/2/21, and 8/2/21 (we know most of the dates when the city posted to social media that the cards were refilled and can also observe spikes in spending when participants learned their cards were refilled, but cannot definitively rule out some spending in advance of the announcements, if the announcements occurred a day or two after the cards were actually credited).

[14] When individuals reported lost cards, their cards were replaced so that they could continue to receive funds going forward, but funds for prior months were not restored.

## 3. Data and methods

In this section we describe our main sources of data for our analysis along with our analytic methods.

*a. Surveys*

Most of our results come from six surveys we administered to our study sample. After the City decided in July 2020 that it was going to allocate the cash cards by lottery, we designed a baseline survey that we administered between September 11, 2020 and 5:25pm on September 17, 2020 (the moment when the lottery occurred). We recruited sample members by texting them at the numbers they provided on their lottery applications and by mailing a postcard to their mailing address. The City supplemented our outreach with a social media campaign asking residents to participate so that the City could better understand the needs of the population during the COVID-19 crisis. Surveys were administered using Qualtrics. Approximately two-thirds of sample members took our surveys themselves by clicking on the link we provided. The remaining third preferred to speak with team members who administered the survey over the telephone. 75 percent of participants took the survey in Spanish, and 25 percent took it in English. The first survey began by obtaining informed consent. Our study procedures were approved by the Harvard University Institutional Review Board.

1936 sample members consented to participate in our study prior to the lottery, and these individuals represent our sample for most of the analysis we present below. Column 6 of Appendix Table A.1 shows the characteristics of our survey sample based on the characteristics reported on the lottery application. Comparing column 6 to column 1, our interview sample appears to be younger than the overall sample – less likely to be age 65+, more likely to have children, and more likely to have a household member working.

The City originally intended to distribute the Chelsea Eats cards immediately after the lottery, but challenges in procuring the cards delayed distribution by two months. We decided to administer a second, "pseudo baseline," survey post lottery but before the first payments were made. This November survey allowed us to collect additional information on the status of the sample prior to receiving the Chelsea Eats cards, stay in touch with sample members, test out possible questions for the final impact survey, and collect some data helpful to City officials on internet access for school children during the pandemic. Moreover, by examining whether treatment and control group responses differ after learning about their lottery status but before the cards were distributed, we can assess whether survey responses are driven primarily by happiness about winning and frustration about losing the lottery versus difference in income from actually receiving the cards.

In December 2020 and February 2021, we administered two very short "pulse surveys." Our primary objective in administering these surveys was to stay in touch with sample members and to make the survey experience pleasant, so that as many sample members as possible would be willing to take our main final outcomes survey. These surveys also allow us to describe how some key outcomes evolved longitudinally over the course of the study period.

In March 2021 we administered a survey that asked respondents to report everything they had eaten in the previous 24 hours. This survey was administered entirely over the phone, and interviewers collected data using the ASA24 dietary assessment tool. Given the complexity of the dietary analysis and the need to involve additional co-authors with nutrition expertise in that part of the project, most of the results from that survey will be reported in a companion paper.

Most of our main study outcomes were collected in a final survey administered in April and May 2022.[15] The final survey contained two parts, each of which required approximately 30 minutes to complete. Respondents could take the parts separately or in a single sitting. Whereas we provided $20 in compensation to respondents for completing each of our prior surveys, we provided $25 in compensation for each half of the final survey -- $50 total. For surveys 2-5 we stopped collecting data when we reached completion rates of around 90% (both to conserve funds and to avoid pestering the more reluctant sample members in advance of the final survey). For the final survey we continued collecting data until we were unable to get any further responses – which resulted in a response rate of 95 percent. We believe we were able to achieve this high response rate for three reasons. First, the City of Chelsea built up a lot of trust with residents during the COVID-19 crisis, and the City's messaging to promote the value of the survey to the community increased people's interest in participating. Second, we built a strong relationship with participants through our approach of staying in touch and getting them accustomed to receiving gift cards from us. Third, we were studying an economically vulnerable population, and the $20 gift cards made a significant difference for many households financially.

Survey questions were drawn from standard instruments whenever possible. For example, food security is measured using the USDA Food Security Survey Module and mental health is assessed using the PHQ-2 and GAD-2. Table A.2 lists survey dates, interview topics, and response rates.

*b. Chelsea Eats card spending*

We have anonymized data on purchases made with all of the Chelsea Eats cards that were distributed, a total of 148,294 transactions amounting to $6 million. The data contain the

[15] 97 percent of the final survey responses were collected in April, so we refer to the final survey as the "April survey" in the rest of the paper.

purchase date, the purchase amount, a "description" that contains the name of the store or vendor from whom the purchase was made as well as the city or town in which the store or vendor is located, and a "category" that is a Visa-provided categorization of the vendor. Our processing of these data consists of two steps:

- First, we aggregated the data to the vendor level. Because different locations for a single chain (e.g., different Dunkin locations) typically have slightly different entries in the "description" field, this involved a detailed review of the data to make sure we successfully included all instances of the same chain or vendor. There are a variety of other circumstances, such as a single restaurant using multiple payment platforms, where we similarly needed to standardize the "description."
- Second, we modified the Visa-provided categorization to better meet our needs. For example, in order to focus on places where food is typically purchased, we present wholesale clubs such as BJ's and Costco as a separate category rather than as a component of retail.

It is important to emphasize that these data reveal where purchases occurred, but not what was purchased. For example, approximately 50 percent of purchases in our sample occur at grocery stores or wholesale clubs. While it is likely that most of these purchases are food, it is, of course, possible to buy many other things at these locations, such as cleaning supplies. A similar issue occurs with "markets & convenience stores." While it is likely that most purchases in that category are food, particularly because the largest share of these dollars were spent at meat and produce markets, it is clearly possible to buy a wide range of products at a Super 99 or a 7-Eleven.

Two types of stores are particularly hard to categorize. The first are stores like Walmart or Target that sell large amounts of both food and non-food products. We mostly followed the original Visa classifications for these stores, putting them under "retail." But there were some Walmart locations that Visa classified as grocery stores, and we maintained that categorization for those locations. The second type of stores that are difficult to categorize are those that Visa describes as "Gas Stations & Convenience Stores." We put all of the gas stations that Visa classified as "Gas Stations & Convenience Stores" under "Transportation" even though some of the purchases at those locations might be for food items or household goods and more appropriately included in the "Markets and Convenience Stores" category.

*c. Gift card spending*

We also have data on spending of the gift cards we provided in appreciation of survey completion. These data are similar in structure to the data from the Chelsea Eats cards, but have the advantage of showing how incremental income is spent for both the treatment group and control group. However, the card vendor for these gift cards was not capable of providing us bulk data on these cards, and we obtained these data by hand, entering each card number individually. Given the time involved, we collected these data for a random sample of our study population.

*d. School attendance data*

We analyzed data on school attendance using administrative records from the Chelsea public schools. School attendance is collected in a standardized way in Massachusetts since it affects state aid (Massachusetts Department of Elementary and Secondary Education, 2020). We matched sample members to school records using parent phone number, parent name, and address for the entire sample of 3615 lottery entrants.

*e. Pre-analysis plan*

We posted a pre-analysis plan to the American Economic Association site on August 17, 2021, six months prior to linking our survey data to treatment status.

*f. Weighting*

Because program applicants received different numbers of lottery tickets, we need to weight the data in order for the treatment and control groups to be balanced. We assign a weight of 1 to each member of the treatment group and a weight of $p/(1-p)$ to each member of the control group where $p$ is the probability of the entrant winning the lottery. This weighting scheme makes both the treatment group and control group match the characteristics of the treatment group. Given that the City was deliberately trying to create a recipient population that was weighted toward those households with greatest need, we think this is the most obvious weighting scheme to use. In practice, this choice does not seem to matter much in our sample. We have re-run our results weighting the data to match the applicant population, rather than the card recipient population, and obtain extremely similar impact estimates.

We calculate $p$ by simulating the lottery 20,000 time and calculating the percentage of times each household wins the lottery. Within groups with the same number of tickets, these simulated probabilities match the actual probabilities of winning the lottery quite closely, further confirmation that the lottery functioned as expected.

*g. Households with multiple lottery entries*

The 3615 lottery applications included 3091 applicants who appear to be the sole applicant from within their household and 524 applicants who belonged to households where two or more members of a household entered the lottery. Multiple lottery entries do not necessarily reflect anything nefarious. In most cases, city employees appear to have simply encountered

more than one member of a household in signing people up for the lottery and then not noticed prior to the lottery that multiple people had entered from the same address. In most cases in which more than one person from the same address won the lottery, the City caught this when distributing the cards and gave out only a single card. In our analysis, we collapse observations to the household level. In cases in which we inadvertently interviewed multiple individuals from the same household because we did not identify all duplicates prior to the baseline survey, we include the sample member who began the survey first. From a weighting standpoint, we assign the household the sum of all of the lottery tickets across the multiple entries in simulating *p*.

*h. Trimming the sample*

On average, households that entered the lottery multiple times had an 87 percent chance of winning the lottery. In our interview sample, this results in us having 151 treatment group members and only 17 control group members among these households with multiple entries. Among households with 11 or more lottery tickets, we have 48 sample members in the treatment group and only 1 in the control group. In order to ensure we have adequate numbers of control group members to estimate experimental contrasts and avoid having control group observations receiving excessive weight, we drop from our sample 139 households (129 treatment and 10 control) that ended up with a probability of winning the lottery that was more than 80 percent. This corresponds to having 8 or more lottery tickets.[16] Figure 1 shows the various steps in building our analysis sample – from the 3615 lottery entries to the approximately 1700 sample members whose data are the basis for our impact estimates.

[16] We have explored the robustness of our results to trimming at 7 or 9 tickets instead and to retaining all observations but capping the weights for members of the control group at 4 (.8/(1-.8)) and appropriately downweighting the corresponding members of the treatment group to maintain sample balance. As would be expected given that the dropped observations are a relatively small proportion of our sample, the impact estimates are quite similar under these alternative approaches.

*i. Regression specification and covariates*

We estimate experimental impacts by regressing outcomes on an indicator variable for being a lottery winner, using the weights described above and robust standard errors. We present results both with and without covariates. Covariates can improve estimation precision and account for chance differences between the treatment and control group in the distribution of pre-random-assignment characteristics. The regression specification is the standard one:

$$Y = Z\pi + X\beta + \epsilon,$$

where Z is an indicator for being a lottery winner, and X is a matrix of covariates measured in the baseline survey. The covariates were all specified in the pre-analysis plan. Missing values for covariates are imputed at sample means. Table A.3 lists the covariates and displays mean values for both the treatment and control group. Consistent with successful randomization, only one P-value out of the 49 differences in the table (pre-Covid income adjusted for household size) is below 0.05.

## 4. Describing the sample and the intervention

In this section, we first present characteristics of our sample at baseline. Then we describe how the status of the control group evolved over the course of the study. Finally, we present data on how the Chelsea Eats cards were spent.

*a. The status of the sample at baseline*

We begin by referring back to appendix tables A.1 and A.3 and noting that our sample is 80 percent female, possibly because women tended to be the family members who were standing in line to receive boxes of free food and therefore were enrolled in the lottery. The sample is 90 percent Latino. Average family size is 3.4. 45 percent of families have a young child, and 60 percent have a school-aged child.

Turning to Table 1, which reports data on household financial situation from our baseline survey collected during the two weeks leading up to the lottery, we see that 80 percent of the sample said they were worse off than a year before, while only 2 percent said they were better off.[17] For many, their financial situation was still deteriorating in September 2020, with 50 percent worse off than one month before and only 7 percent better off. 82 percent of households reported experiencing a financial hardship in the past year, with lost jobs (54 percent), reduced hours (34 percent), spouse lost job (22 percent), spouse had reduced hours (18 percent), a health emergency (20 percent), and increased prices for food and other essential items (45 percent) the most common contributors. Indicative of the horrors of the initial wave of COVID-19, 4.8 percent reported that the death of a household member had contributed to their financial hardship.

Among the 87 percent of sample members who experienced a time in the past year when they did not meet all of their essential expenses, 94 percent were still struggling to meet essential expenses in September. Average monthly income was down 38 percent compared to pre-COVID-19. 71 percent had bills or expenses they had been unable to pay in August, with an average income gap of $1357. Averaging in zeros for those who did not have unpaid expenses, the average shortfall was $950, very similar to the $1080 decline in average monthly income over the COVID-19 period. 39 percent were behind in their rent, and 45 percent were behind in their utility bills. Approximately 20 percent had had each of three types of utilities disconnected in the past four months because of not paying bills, and 21 percent had delayed getting medical care.

[17] To be as representative as possible of Chelsea Eats applicants, our analysis in this subsample uses data from all 1936 sample members who enrolled in our study (without trimming those with more than 8 lottery tickets). 116 of the 1936 enrolled but did not answer any further questions in the baseline survey.

Table 2 shows that the financial struggles facing this sample carried over into food security and health. The first question presented in the table is the standard one-question assessment of food security. At the time of the baseline survey only 12 percent of the sample had enough of the kinds of food they wanted. 38 percent had enough food, but not always the kinds of food they wanted. 39 percent sometimes did not have enough to eat, and 11 percent often did not have enough to eat. This is an extraordinary level of food insecurity for a U.S. population. For comparison, the Massachusetts-wide numbers during the same month as measured by the U.S. Census Pulse Survey were 67.0 percent, 25.6 percent, 5.5 percent, and 1.8 percent, respectively. So our Chelsea sample, largely recruited from people standing in line for free food, had seven times as many people who sometimes or often did not have enough to eat.

We also administered the Six-Item Short Form USDA Household Food Security survey. 40 percent of our sample scored a 5 or 6, which is classified as very low food security. Nationwide, only 3.9 percent of Americans experienced very low food security in 2020 (Coleman-Jensen et al, 2022). A further 44 percent of our sample scored between 2 and 4, which is classified as low food security. Nationwide only 6.6 percent of Americans fell in that category.

At baseline, 42 percent of our sample reported fair or poor health, more than double the national average for Latinos in a similar age range. The percent scoring 13+ on the K6 measure of psychological distress was 35 percent; in national samples during non-COVID years, the typical average is around 4 percent. At baseline, 32.5 percent of respondents reported that they or a family member had experienced COVID-19, an extremely high number at that point in time.

Table 3 describes the work situation of the sample at baseline. 36 percent were working for pay, 46 percent were unemployed and looking for work, and 18 percent were unemployed

and not looking for work. Among those working, median hours worked per week were 30. Consistent with an economy that had only barely started to reopen, 64 percent of workers preferred more hours of work, and only 9 percent preferred fewer. Common reasons for not working were unavailability of work, own illness or disability, lack of childcare, personal or family responsibilities, and preferring to stay home with children. Table A.4 shows occupations of sample members. 30 percent were employed in the food industry, 26 percent were employed in housekeeping or janitorial work, and 11 percent were employed in customer service.

Tables A.5, A.6, and A.7 present additional descriptive information about the sample, mostly collected during the November survey that occurred after the lottery but before the Chelsea Eats cards were distributed. Here we highlight a few tabulations that will be helpful in interpreting later results. First, while we did not directly collect information on legal immigration status, some of the measures of government benefit receipt in table A.5 can help in assessing how many sample members are likely to be undocumented. In particular, 26 percent of the sample reports receiving health insurance from MassHealth Limited, the Massachusetts health plan for undocumented residents. In addition, we see that despite the significant economic distress in this sample, only 47 percent of sample members are receiving SNAP, indicative of many being ineligible because of immigration status, and 38 percent of SNAP cases are children-only cases, which may represent households in which the children were born in the United States to undocumented parents. Turning to table A.6, we see that 64 percent of households received free groceries or a free meal in the past seven days and that among households receiving free meals, 55 percent obtained half or more of their household's food from free sources. We also see that in September, 9 percent were receiving money from relatives or friends, 27 percent were sending money to relatives or friends living outside the U.S., and 8 percent were sending money

to relatives or friends inside the U.S. Table A.7 shows that 56 percent of sample members had a bank account, and the average balance among those with an account was $1132. Among the 30 percent of sample members that had credit cards, the average debt was $5025.

*b. How the situation for the control group evolved*

Measuring the trajectory of the control group across our surveys can help to describe how the situation in Chelsea changed over the course of the program as the economy gradually reopened and other relief efforts expanded, thereby providing context for the experimental impacts we present in the next section of the paper. In this part of analysis, we present results for the trimmed portion of our sample so as to match the control group sample that will be analyzed in the experimental impacts. We further limit our sample for each question to individuals who answered that question in all of the surveys, so that our results come from a balanced panel and are not the result of variations in which sample members responded to the survey in different months. In practice, this balanced panel restriction does not matter. Results are nearly identical for the entire set of control group respondents to each survey which reflects both our high response rates and that the remaining survey and item nonresponse does not appear to be correlated with these outcomes.

In the first panel of table 4, we see responses to the question of whether the household is better off or worse off financially than last month. This question was asked in all five of the regular surveys (but not in the ASA24 food recall survey). The results show that prior to the final survey, more households experienced things getting worse than getting better, but the proportion answering "worse" mostly declines (the exception is the December survey) and the percentage answering "the same" increases. Only the final survey in April shows more households getting better than getting worse.

In general, the perception we had as we interacted with these families over the September 2020 to Winter 2021 period was that as the economy gradually reopened, the gap between people's needs and their incomes narrowed but did not disappear, meaning that families were still getting further behind in what they owed in rent and utilities, but at a slower rate. The respondents' answers to "Were you able to pay your full rent/mortgage last month?" are consistent with this interpretation. The percentage answering "yes" declines from 68 percent in the November survey to 63 percent in December and to 59 percent in February. Only in the final April survey do we see an improvement – to 71 percent.

Similarly, the percentage who say they are "highly confident" or "moderately confident" that they will be able to pay their rent next month falls from 32 percent in November to 23 percent in December, before rebounding to 31 percent in February and 38 percent in April. The percentage of respondents working for pay falls from 45 percent in the November survey to 39 in December, before returning to 44 percent in April, showing that even as the economy continued to reopen, these respondents, highly concentrated in the cleaning and restaurant/food preparation sectors, were still struggling to find work.

The perception that things were finally somewhat better for the control group in April 2021 than in September 2020 is confirmed by the responses to whether the household had any unpaid bills or expenses. In April, 58 percent of households said they had unpaid bills or expenses, down from 73 percent in September.

In contrast to the financial situation and rent questions which show a pattern that continues to get worse before it gets better, our measures of food security show steady improvement. The percent of respondents saying their household often does not have enough to eat declines from 11 percent in September to 6 percent in November, 5 percent in December, and

3 percent in April. The percentage saying their family sometimes does not have enough to eat similarly shows a steady decline, from 40 percent in September to 17 percent in April. The percentage of respondents who say their children "often" or "sometimes" do not get enough food because the household cannot afford food falls from 56 percent in September to 51 percent in November, 48 percent in February, and 40 percent in April. The Chelsea community was continuing to ramp up food relief efforts throughout the Fall of 2020, so it is possible we see steadier improvement in this outcome as the result of those efforts. Consistent with this interpretation, use of free food in the past week remains high throughout the study for the control group at 62 percent in November, 65 percent in December, 61 percent in February, and 68 percent in April. The February decline is likely to be weather related; when we asked sample members why they did not use free food sites, weather was one of the most common responses.

We assessed depression and anxiety using the two-question and PHQ-2 and GAD-2 measures in three of our surveys. The percentage of the control group who met the threshold for likely depression rose from 24 percent in November to 27 in February before falling to 22 percent in April. The percentage who met the threshold for anxiety disorder was steady at 33 percent in both November and February before decreasing to 26 percent in April.

*c. Spending of the Chelsea Eats cards*

Program participants could use their cards at any store or vendor that accepted Visa.[18] We are able to analyze anonymized data showing where cards were spent and have categorized the spending by type of establishment. Figure 2 shows aggregate daily purchases using the Chelsea Eats cards. Spending on the cards spikes around each date when the cards were refilled,

[18] It was not possible to withdraw card balances as cash.

consistent with the population being financially constrained.[19] Table 6 shows spending by category for each "month" of the program, with the months defined by the dates on which the cards were restocked.[20] The first four rows of the table represent spending at locations at which food is the main product being sold – grocery stores, wholesale clubs, markets and convenience stores, and restaurants.[21] In the first month of the program, 60.0 percent of spending was at grocery stores and another 22.5 percent was at wholesale clubs, markets, or restaurants, so 82.5 percent was at places in which food was the main product. Most of the remaining spending was at retail stores. As the program progressed the share spent on food declined and the share spent on retail increased. The pattern is likely to result from the combination of two factors. First, as the economic situation of treatment group members improved and as other food relief efforts in the city expanded, households may have increased their expenditures on non-food items. Second, because the program was marketed as "Chelsea Eats," it may have taken time for some sample members to realize it was possible to spend their cards on non-food purchases.

Over the full nine months of the program, $6 million of spending occurred. 65 percent of the $6 million was spent at places in which food was the primary product (39 percent at grocery stores, 11 percent at wholesale clubs, 5 percent at markets/convenience stores, 10 percent at restaurants). 27 percent was spent at retail stores.

[19] Other models such as sample members having a high discount rate or concern that the funds might disappear or the cards might get lost could also produce this pattern.
[20] Card balances could be carried over across months, and a handful of respondents told us they were holding onto to their cards for emergency expenses rather than spending them.
[21] In Liebman et al (2021), we present additional detail on the first $2 million of card spending, including a breakdown of the largest vendors within each category. The local Market Basket grocery store was the location where the greatest amount of purchases occurred -- 33 percent of overall purchases in that time period. The largest amount of wholesale club purchases occurred at BJ's and Costco. The largest amount of purchases at markets and convenience stores was at La Economica Meat, Vincentinos Meat Market, and Vincentinos Produce, all local Chelsea vendors. Dukin, McDonald's, Uber Eats, and Burger King received the largest amounts of restaurant spending, followed by four local restaurants, Broadway House of Pizza, Delicias Salvadorenas, Taqueria El Charrito and Pollo Campero. In the retail category, Walmart, Target, Burlington Stores, Amazon, and Family Dollar received the greatest amount of spending.

## 5. Experimental impacts: primary outcomes

In our pre-analysis plan we pre-specified five outcomes that would be evaluated to assess the overall success of the intervention: financial well-being, food insecurity, food satisfaction, health/mental health, and K-12 school attendance. For the first four, we pre-specified indices that would aggregate results over multiple outcomes in those domains. We also specified two additional key outcomes, employment and diet quality, but stated that those two would not be included in the omnibus test of overall success of the intervention because it was less clear what result would represent success.[22] In this section, we first describe the overall results and then go into further detail within each domain, presenting the subcomponents of the indices and other related outcomes.[23]

### a. *Overall impacts*

Table 7 presents results for our five primary outcomes and for an overall summary index. The first four columns show results without covariates. The final two columns show results with covariates. The first row shows that the treatment group scored approximately 3 percentage points higher on the financial well-being index, a result that is statistically significant at the 5 percent level (p-value for unadjusted difference, .012; p-value for the regression-adjusted difference, .034). This index can be interpreted as the average fraction of questions in the index that received a positive response, so the control group value of .302 implies that the average person in the control group responded affirmatively to 30 percent of the financial well-being questions.

[22] While there are benefits to the economy of increased output, if the extra income permitted people to delay going back to unsafe jobs during the COVID pandemic or to stay home and supervise their children in online learning, then a decline in employment could represent a beneficial outcome. Similarly, if the extra income permits people to switch from eating rice and beans to eating beef and pork, is that a good outcome or bad outcome from a diet quality perspective?

[23] Analysis of the 24-hour food recall data is ongoing. While we present some results on diet quality from the April survey in this draft, we will be adding additional results from the recall data in a subsequent draft.

The percentage of households reaching the threshold for food-insecure on the USDA 18-item food security scale is presented in row two. While the point estimates show a decline of 3.9 percentage points in food insecurity in the unadjusted results and a decline of 1.5 percentage points in the covariate-adjusted results, these differences are not statistically significant.

The third row shows results for the food satisfaction index. There was a statistically significant increase of 5 percentage points in the unadjusted results and 4 percentage point in the adjusted results from a base of 24 percent. The fourth and fifth row show results from two measures of potential downstream impacts, physical/mental health and number of days a child was absent from school. Neither show any impact of the program.

By constructing summary indices and pre-specifying five primary outcomes, we reduce the number of study endpoints and the chance of type I error. However, it is still helpful to assess how likely it is that we would obtain the two statistically significant results that we obtain under the null hypothesis that the experiment had no effects. To do so, we create a summary index of our first four primary outcomes by standardizing them and aligning their signs so that a positive difference represents an improvement in the outcome.[24] The sixth row of the table shows that the there was a .11 of a standard deviation gain for the treatment group in the unadjusted results and a .07 of a standard deviation gain in the adjusted results, with P-values of .003 and .044 respectively – consistent with the experiment having an overall affect.[25]

[24] We are unable to include the school attendance result in the overall index both because the sample is different and because we can only access those data on site at the school department.

[25] We have also applied the Westfall and Young (1993, Algorithm 2.8) stepdown resampling method to adjust the per comparison p-values. Conceptually, the Westfall-Young method multiplies the lowest P-value by the number of tests and then, for the second lowest P-value, it assesses how likely it is under the null that the second lowest P-value will be below the estimated P-value, and so on for the remaining P-values. Simulation-based methods of estimating P-values are used to account for correlations across measures. The P-values we show in Table 7 are asymptotic standard errors using Stata's default for robust standard errors (HC1). A recent literature argues that it is preferable to use randomization inference (Young, 2019) or robust HC3 standard errors (Simonsohn, 2021). When we redo our estimates with HC3 standard errors, the P-values in our unadjusted results do not change at all, and the regression adjusted P-values rise from .034 to .038 for the financial well-being index and from .012 to .014 for the food

*b. Impacts on financial well-being*

Table 8 shows the components of the financial well-being index. Our index puts one-third of the weight on each of three concepts: a general sense of wellbeing as measured by whether the respondent says he or she is better off financially than six months ago, the ability to pay bills and expenses, and the ability to pay for a $300 emergency expense with money on hand. We further subdivide the ability to pay bills category, putting half of the weight on an overall measure of having any expenses the respondent was unable to pay and splitting the remaining half across the ability to pay utilities, rent, and healthcare costs. The first column of the table shows the index weights for each measure. The only component of the index that shows a statistically significant impact on its own is the measure of whether the respondent is better off financially than six months ago – which rises by 6.6 percentage points from a base of 20.5 percent. The other two components with nonnegligible positive point estimates are the ability to pay full utilities and ability to pay for a $300 emergency expense.

The findings in this table, along with food and clothing expenditure results below, are consistent with a pattern in which card recipients used the cards to make incremental purchases of food and other necessities, but did not have enough left over to fully pay rent and utility bills. The relatively high rate of health insurance coverage in the sample and the related low baseline

satisfaction index. When we apply randomization inference, we obtain much lower P-values than with our asymptotic regression estimates, a result we don't currently understand given that our sample size is fairly large. So to be conservative, we are not reporting the randomization inference P-values or Westfall-Young simulations that are derived from them in this draft. Applying the Westfall-Young logic to the P-values in table 7, the unadjusted results are clearly significant at the 5 percent level even accounting for multiple testing, as the food satisfaction would only reach .01 after being multiplied by 5, and the probability of having two per comparison P-values below .012 is quite low (even with the two variables having a correlation of .47). For the regression adjusted results, the food satisfaction P-value would rise to .06 after adjustment. While the probability of having two P-values below .034 in five tests is quite low, the Westfall-Young procedure imposes monotonicity on the P-values and would also adjust the financial well-being P-value to be .06. So overall, it seems quite likely that the experiment had an impact, but in the regression-adjusted results there is a bit more than a 1 in 20 chance of the financial well-being or food satisfaction result being a false rejection of the null.

rate of being unable to obtain health care because of costs may explain the lack of an experimental impact on delaying health care or prescriptions. Appendix table A.8 shows experimental impacts for secondary outcomes in the financial well-being domain. There is a consistent pattern in these results with the financial situation question showing statistically significant treatment-control differences in each month, while the rent-related questions show no differences.

*c. Impacts on food security*

Before turning to the food security and food satisfaction results, it is helpful to first review experimental impacts on food-related secondary outcomes that provide context for the primary outcomes. Table 9 shows a variety of food consumption measures. In particular, in the April 2020 survey we collected data on household's usual weekly expenditures at grocery stores, convenience stores, restaurants, and food delivery. The first row of Table 9 shows that, on average, the treatment group spent $38 more per week than the control group on food. To put this in context, the average Chelsea Eats monthly card amount was $350 and in April 2020, 63 percent of card spending was at food-related stores. Taking these numbers at face value, this implies that 75 percent of card spending represented incremental spending, and 25 percent was replacing spending on food that would otherwise have occurred.[26]

Table 9 also displays data on the percentage of households that ate various types of food during the past week. The treatment group was more likely to have consumed fresh vegetables, fresh meat, fish, and processed meat. There was no difference on legumes, possibly because the additional financial resources provide by the Chelsea Eats card caused families to substitute out

[26] Multiplying $38 x 4.35 weeks to turn it into a monthly amount yields $163 of incremental food expenditures by the treatment group. 63 percent of $350 is $221.

of legumes and toward more expensive proteins, or because families could easily obtain legumes at free food sites.

Finally, the table shows that treatment group households were *more* likely to receive free food boxes than members of the control group, but less likely to receive the majority of the household's food from such sites. We have two hypotheses for why treatment group households were more likely to receive free food. First, free food may be more valuable if the food can be supplemented with other items. For example, free potatoes and onions may be more desirable if one can also afford meat and cooking oil. Second, card winners may have felt more socially connected to and trusting of the city's food relief networks because of receiving the cards.

We turn now to Table 10 that contains results on food security. In the December survey that was administered two to five weeks after the cash cards were first credited, we administered both the standard one-question household food security measure and the one-question child food security measure. On the household question, the percentage of respondents who said the household "sometimes" or "often" did not have enough to eat was 10 percentage points lower for the treatment group, with a control group mean of 32 percent. The P-value was .000 to three decimal places for both the without-covariates and with-covariates result. On the child food security question, we find that the treatment group was 5.1 percentage points in the results without covariates and 4.3 percentage points in the regression-adjusted results less likely to report that their children "sometimes" or "often" did not have enough to eat because the family could not afford food, down from 53 percent in the control group. The P-values for these results were .093 in the results without covariates and .119 in the results with covariates. In the February 2021 survey, we asked only the child food security question. The percentage of respondents who said their children "sometimes" or "often" did not have enough food to eat was

7.2 percentage points lower (P-value of .025) in the results without covariates and 4.6 percentage points lower in the results with covariates (P-value of .099).

Table 10 shows four measures of food security from the April 2021 survey. The first is the percent who are food secure using the full USDA 18-item scale. That result is simply reprinted from the overall summary (Table 7). We also show the continuous measure of food security from the USDA 18-item scale which converts responses from families with and without children to a comparable 10-point range as well as the same one-question items for households and children collected in the earlier months. None of these three measures show significant treatment-control differences in April. Taken as a whole, these results suggest that the intervention likely reduced food insecurity in its initial months, but by April 2021 there is no evidence of an experimental impact on food security.

*d. Impacts on food satisfaction*

Our food satisfaction index is an equally weighted summary of three measures from the April 2020 survey: the percentage of households who were "very satisfied" with the family's food situation, the percentage of families reporting that their food situation was "better" than six months prior, and the percentage answering "enough of the kinds of food we want" to the one-question USDA household food security item. Table 11 shows that the point estimates are higher for the treatment group than the control group for all three measures and that the "family food situation is better than six months ago" measure has the lowest P-values.

*e. Impacts on physical and mental health*

Table 12 shows the components of our health index. Fifty percent of the index weight is on the percentage of households answering that their health was "fair" or "poor" to the standard one-question general health measure. The other fifty percent of the index weight is on whether

the respondent scored a 3 or above on either the GAD-2 general anxiety disorder screen or the PHQ-2 depression screen. The treatment-control difference was not statistically significant for either measure.

*f. Impacts on employment and hours of work*

Table 13 shows the impacts of the intervention on employment and hours of work. We show results both for the respondent and for a household measure that for married couples also includes the respondent's spouse. The point estimates are all in the direction of greater employment and hours for the treatment group which received the Chelsea Eats cards, but none of the treatment-control differences are statistically significant.

## 6. Experimental impacts: secondary outcomes

In addition to the secondary outcomes already presented on financial well-being and food consumption, we collected information on a variety of other outcomes. Table 14 presents results for a selection of these additional outcomes. The point estimates suggest that the treatment group was more likely to pay for child care, had greater spending on clothing, had more money in cash and in bank accounts and was more likely to have had COVID-19, but none of these results are statistically significant at the 5 percent level.

Two of the results in the table are statistically significant. Members of the treatment group were 4 percentage points less likely to have moved homes since November 2020, a move rate that was nearly one-third lower than the 12.4 percent rate in the control group. 3.4 percent of female treatment group respondents were pregnant at the time of the April survey compared to 1.3 percent of those in the control group. These results could be consistent with the overall improvement in the financial situation of the treatment group if the decline in moves was the result of not being evicted because of non-payment of rent and the increase in pregnancy was the

result of feeling financially secure enough to add a family member. However, we urge caution in interpreting these results for two reasons. First, these results could be due to type I error given that they were selected from an overall list of approximately a dozen secondary outcomes. Second, while the results have the sign we were expecting when we decided to include these measures in our survey instrument, if the results had come out with the opposite sign there would be plausible explanations for such results.[27]

## 7. Conclusion

The Chelsea Eats program was implemented in the midst of a food crisis in the City of Chelsea, with thousands of families struggling to obtain enough food to eat and the City strained by running a food distribution program. The program aimed to boost food access, satisfaction, and choice, while lowering overall financial distress. Our results show that the program largely achieved its goals. 65 percent of card spending was at grocery stores and other locations that sold food. Food expenditures and consumption were higher among households that won the lottery and received cards. Those who received the cards were more likely to say that their household's food situation and overall financial situation had improved. The program did not, however, have impacts on our two primary downstream outcomes: health (including mental health) and school attendance.

While the percentage of households in our study sample that were food insecure declined over the study period as the economy reopened after the COVID-19 shutdown and food relief efforts expanded, there was still substantial unmet financial need and high rates of food insecurity in both the treatment group and control group at the end of our study period. It would

[27] For example, had there been an increase in moves for the treatment group, it would be plausible to argue that the greater financial resources enabled people to leave overcrowded conditions.

likely have required providing households with more than $400 per month to bring down these rates further.

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## Figure 1: Experimental Samples

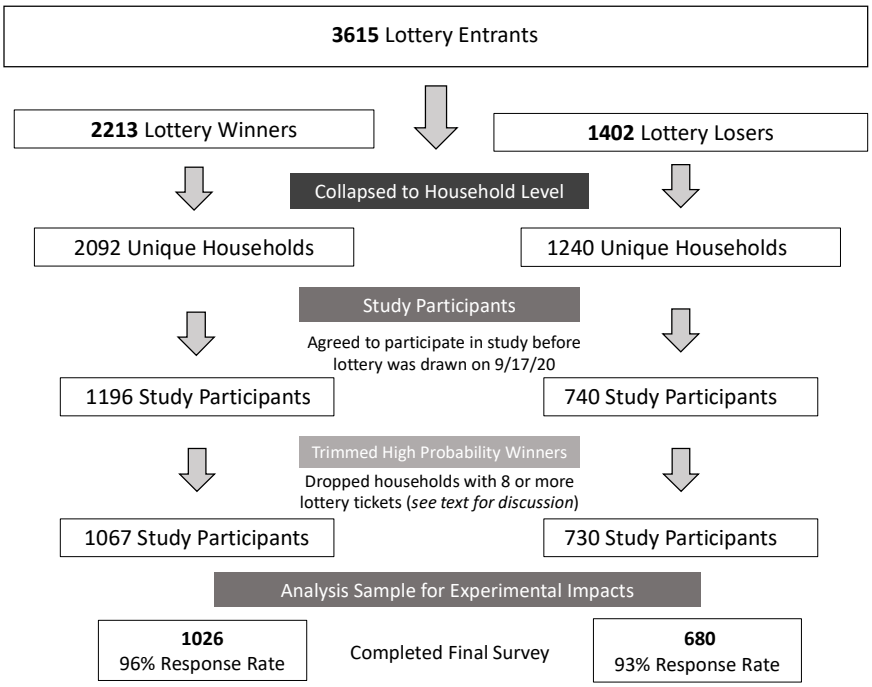


**Figure 2: Daily Purchases from Chelsea Eats Cards**

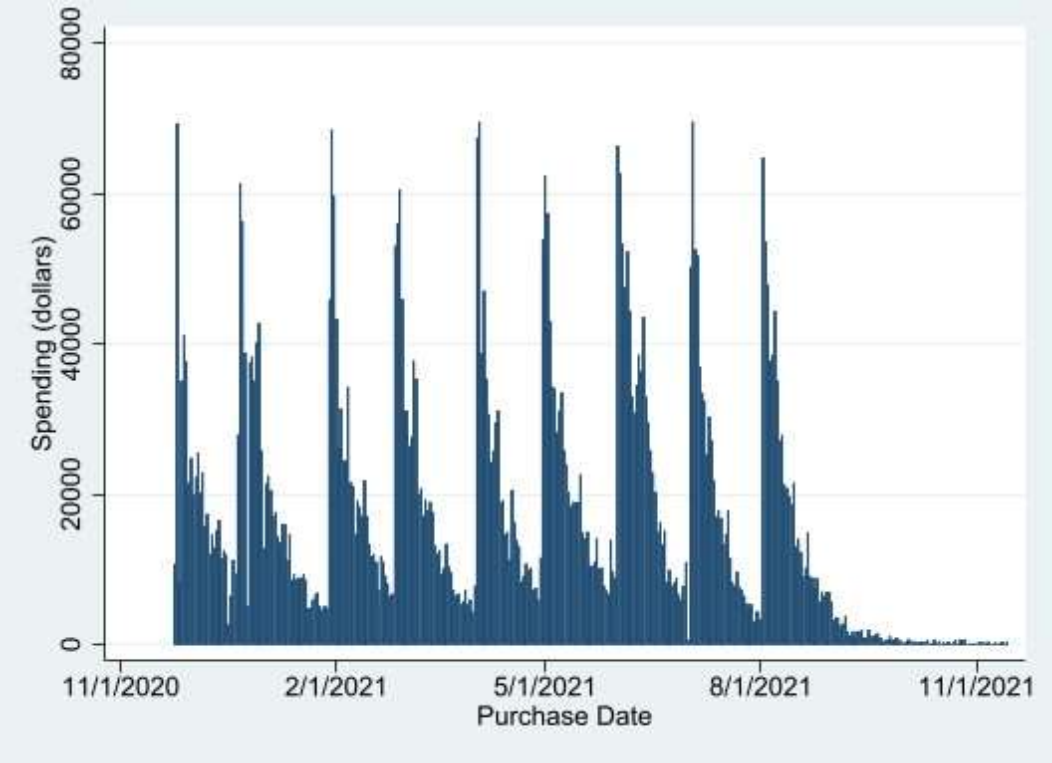


**Table 1: Financial Situation**

| Baseline survey (September 2020) | |
|---|---|
| **Would you say that you are better off or worse off financially than you were a year ago? [N=1820]** | |
| Better | 2.4% |
| Worse | 80.0% |
| About the same | 17.6% |
| **Would you say that you are better off or worse off financially than you were a month ago? [N= 1796]** | |
| Better | 6.7% |
| Worse | 49.8% |
| About the same | 43.4% |
| **Over the past year, have you or members of your household experienced any financial hardship? [N= 1803]** | |
| Yes | 82.3% |
| No | 17.7% |
| **Which of the following contributed to your financial hardship [N= 1486, asked only if answer to preceding question was "yes," (multiple responses permitted)]** | |
| Lost job | 54.3% |
| Work hours and/or pay reduced | 34.3% |
| Spouse lost job | 22.3% |
| Spouse work hours and/or pay reduced | 18.4% |
| Business I owned had financial difficulties | 1.9% |
| Separation/divorce from spouse/partner | 4.2% |
| A health emergency | 19.6% |
| Loss/reduction in government benefits | 8.2% |
| Received foreclosure/eviction notice | 2.2% |
| Death of household member | 4.8% |
| Receiving fewer meals at school or from community groups | 5.5% |
| Price increases for food and other essential items | 45.0% |
| Other | 6.9% |

**Table 1: Financial Situation, continued**

| Baseline survey (September 2020) | | |
|---|---|---|
| **During the past 12 months, has there been a time when you did not meet all of your essential expenses? [N= 1787]** | | |
| Often (almost every month) | | 27.8% |
| Sometimes (once every few months) | | 42.5% |
| Rarely (once or twice a year) | | 16.7% |
| Never | | 13.0% |
| **Are you currently struggling to meet your essential expenses? [N= 1560, not asked if answer to preceding question was "never"]** | | |
| Yes | | 93.8% |
| No | | 6.2% |
| **Monthly Income** | | |
| Before Covid [N= 1737] | $ | 2,820 |
| Now (Aug 2020) [N= 1712] | $ | 1,739 |
| **Did your household have any bills, expenses, or needs that you weren't able to pay for in August 2020? [N=1721]** | | |
| Yes | | 70.8% |
| No | | 29.2% |
| How much more money would you have needed in August to make ends meet? | $ | 1,357 |
| **Are you currently behind on your rent or mortgage? [N= 1708]** | | |
| Yes | | 39.0% |
| No | | 61.0% |
| **Are you currently behind on your utility bills? [N= 1662]** | | |
| Yes | | 45.0% |
| No | | 55.0% |
| **In the past four months, were any of these services disconnected because you were not paying the full amount of the bills? [N= 1687]** *(percent "yes" response)* | | |
| Gas, oil, electric | | 22.8% |
| Mobile/cell phone service | | 20.5% |
| Internet/TV service | | 22.7% |
| **During the past 4 months, have you delayed getting medical care because of the cost? [N= 1685]** | | |
| Yes | | 20.8% |
| No | | 79.2% |

Source: Authors' calculations from baseline survey.

**Table 2: Food Security & Health**

| Baseline Survey (September 2020) | |
|---|---|
| **Which of these statements best describes the food eaten in your household in the last four months? [N=1689]** | |
| Enough of the kinds of food we want | 12.2% |
| Enough but not always the kinds of food we want to eat | 37.8% |
| Sometimes not enough to eat | 39.2% |
| Often not enough to eat | 10.8% |
| **My children were not eating enough because we couldn't afford enough food [N= 1341, asked only of households with children]** | |
| Often true | 13.4% |
| Sometimes true | 41.0% |
| Never true | 45.6% |
| **USDA Household Food Security Survey: Six-Item Short Form [N= 1599]** | |
| 0 -- high food security | 8.9% |
| 1 -- marginal food security | 7.3% |
| 2 -- low food security | 22.6% |
| 3 -- low food security | 12.5% |
| 4 -- low food security | 8.9% |
| 5 -- very low food security | 12.2% |
| 6 -- very low food security | 27.5% |
| **In general, would you say your health is... [N= 1671]** | |
| Excellent | 8.1% |
| Very good | 17.5% |
| Good | 32.8% |
| Fair | 34.5% |
| Poor | 7.1% |
| **Kessler Psychological Distress Scale (K6) [N= 1661]** | |
| None/low mental distress (0-4) | 15.8% |
| Moderate mental distress (5-12) | 49.2% |
| Severe mental illness (13+) | 35.0% |
| **Respondent or family member has had Covid? [N= 1620]** | |
| Yes | 32.5% |
| No | 67.5% |

Source: Authors' calculations from baseline survey.

**Table 3: Work Situation**

| Baseline survey (September 2020) | |
|---|---|
| **Which of the following best describes your current situation? [N= 1587]** | |
| Working for pay | 36.1% |
| Unemployed and looking for work | 45.5% |
| Unemployed and not looking for work | 18.4% |
| **How many hours a week do you usually work at this job [if working for pay]? [N= 541]** | |
| 0 - 19 hours | 17.4% |
| 20 - 25 hours | 22.6% |
| 26 - 35 hours | 33.8% |
| 36 - 40 hours | 24.0% |
| 41+ | 2.2% |
| **Would you prefer to work more hours at this job, fewer hours, or no change [if working for pay]? [N= 541]** | |
| More | 64.2% |
| Fewer | 8.7% |
| No Change | 27.1% |
| **Why are you not working for pay? [N= 986]** | |
| Suitable work is unavailable or hard to find | 50.9% |
| Lack necessary skills or qualifications | 10.8% |
| Lack transportation to or from work | 8.8% |
| Own illness or disability | 22.2% |
| Unable to find good childcare | 16.1% |
| Caring for elderly relative(s) | 2.6% |
| Attending school | 3.4% |
| Prefer to stay at home with children | 18.1% |
| Personal or family responsibilities | 14.9% |
| Gave up looking for work | 6.1% |

Source: Authors' calculations from baseline survey.

**Table 4: Evolution of Control Group - Financial Situation**

| | Sep. 20 | Nov. 20 | Dec. 20 | Feb. 21 | Apr. 21 |
|---|---|---|---|---|---|
| **Would you say that you (and your family living there) are better off or worse off financially than you were a month ago?** | | | | | |
| Better | 6.8% | 7.7% | 5.3% | 8.7% | 16.7% |
| Worse | 47.4% | 32.5% | 38.5% | 29.1% | 13.4% |
| Same | 45.9% | 59.8% | 56.2% | 62.2% | 69.9% |
| N | 532 | 532 | 532 | 532 | 532 |
| **Were you able to pay your full rent/mortgage this month?** | | | | | |
| Yes | | 68.4% | 63.0% | 59.1% | 71.4% |
| No | | 31.6% | 37.0% | 40.9% | 28.7% |
| N | | 548 | 548 | 548 | 548 |
| **How confident are you that you will be able to make your next rent/mortgage payment?** | | | | | |
| Highly confident | | 14.3% | 9.8% | 12.5% | 15.0% |
| Moderately confident | | 17.5% | 13.6% | 18.4% | 23.0% |
| Slightly confident | | 32.5% | 37.3% | 35.5% | 33.2% |
| Not at all confident | | 35.0% | 38.6% | 32.1% | 28.2% |
| Payment will be deferred | | 0.7% | 0.7% | 1.4% | 0.5% |
| N | | 560 | 560 | 560 | 560 |
| **Did you do any work for pay at a job or business last week?** | | | | | |
| Yes | | 44.8% | 39.1% | | 43.7% |
| No | | 39.8% | 49.1% | | 43.7% |
| Retired | | 3.1% | 2.3% | | 3.2% |
| Disabled | | 4.8% | 4.1% | | 4.8% |
| Unable to work | | 7.5% | 5.4% | | 4.5% |
| N | | 558 | 558 | | 558 |
| **Did your household have any bills, expenses or needs that you weren't able to pay for last month?** | | | | | |
| Yes | 72.8% | | | | 57.8% |
| No | 27.2% | | | | 42.2% |
| N | 595 | | | | 595 |

Source: Authors' calculations from September 2020, November 2020, December 2020, February 2021, and April 2021 surveys.

**Table 5: Evolution of Control Group - Food Security and Health**

| | Sep. 20 | Nov. 20 | Dec. 20 | Feb. 21 | Apr. 21 |
|---|---|---|---|---|---|
| **Which of these statements best describes the food eaten in your household?** | | | | | |
| Enough of the kinds of food we want | 11.2% | 17.7% | 17.9% | | 17.5% |
| Enough but not always the kinds of food we want | 38.7% | 47.6% | 50.6% | | 62.4% |
| Sometimes not enough to eat | 39.6% | 28.3% | 26.9% | | 17.3% |
| Often not enough to eat | 10.6% | 6.4% | 4.6% | | 2.8% |
| N | 502 | 502 | 502 | | 502 |
| **My children were not eating enough because we couldn't afford enough food.** | | | | | |
| Often true | 13.6% | 5.1% | 7.3% | 7.3% | 6.3% |
| Sometimes true | 42.3% | 46.1% | 45.1% | 41.0% | 33.8% |
| Never true | 44.2% | 48.9% | 47.6% | 51.7% | 59.9% |
| N | 317 | 317 | 317 | 317 | 317 |
| **During the last 7 days, did you or anyone else in your household get free groceries or a free meal?** | | | | | |
| Yes | | 61.6% | 65.3% | 61.2% | 67.5% |
| No | | 38.4% | 34.7% | 38.8% | 32.5% |
| N | | 544 | 544 | 544 | 544 |
| **Screened positive for...** | | | | | |
| Depression [N= 520] | | 24.2% | | 27.1% | 22.1% |
| Anxiety [N= 554] | | 33.0% | | 33.2% | 26.0% |

Source: Authors' calculations from September 2020, November 2020, December 2020, February 2021 and April 2021 surveys. The length of the reference period for the household and children food security questions varied across the surveys. The September 2020 survey asked about the last four months, November 2020, December 2020 and February 2021 asked about last week and April 2021 asked about last month.

**Table 6: Chelsea Eats Card Spending by Category**

| Spending Category<br>*(% of dollars spent)* | Nov. 25 - Dec. 21, 2020 | Dec. 22, 2020 - Jan. 28, 2021 | Jan. 29 - Feb. 25, 2021 | Feb. 26 - Apr. 1, 2021 | Apr. 2 - Apr. 29, 2021 | Apr. 30 - May 31, 2021 | Jun. 1 - Jul. 1, 2021 | Jul. 2 - Aug. 1, 2021 | Aug. 2 - Nov. 14, 2021 | Total |
|---|---|---|---|---|---|---|---|---|---|---|
| Grocery stores | 60.0 | 50.0 | 44.5 | 40.8 | 35.6 | 33.6 | 31.4 | 30.8 | 30.0 | 39.1 |
| Wholesale clubs | 12.5 | 11.1 | 11.7 | 11.2 | 11.6 | 10.8 | 11.7 | 9.9 | 10.4 | 11.2 |
| Markets & Convenience Stores | 5.5 | 5.8 | 5.6 | 5.3 | 5.2 | 5.1 | 4.6 | 5.0 | 5.1 | 5.2 |
| Restaurants & Restaurant Delivery | 4.5 | 6.6 | 8.4 | 9.4 | 10.4 | 11.4 | 9.8 | 12.4 | 11.6 | 9.5 |
| Retail | 14.0 | 21.5 | 22.2 | 24.8 | 28.1 | 29.8 | 31.9 | 31.9 | 31.8 | 26.6 |
| Professional Services | 3.0 | 4.4 | 6.3 | 7.2 | 7.5 | 7.8 | 8.8 | 8.3 | 9.5 | 7.1 |
| Travel & Transportation | 0.4 | 0.7 | 1.2 | 1.4 | 1.6 | 1.5 | 1.9 | 1.8 | 1.7 | 1.4 |
| **Transactions** | **11,611** | **16,298** | **14,954** | **17,565** | **15,748** | **18,229** | **18,704** | **16,246** | **18,937** | **148,294** |
| **Dollars** | **$557,736** | **$690,787** | **$617,460** | **$689,747** | **$623,094** | **$710,541** | **$810,204** | **$619,116** | **$684,060** | **$6,002,745** |

**Table 7: Primary Outcomes from April 2021 Final Survey**

| | Without covariates | | | | | With covariates | | |
|---|---|---|---|---|---|---|---|---|
| | Treatment (1) | Control (2) | Difference (3) | P-value (4) | | Difference (5) | P-value (6) | |
| Financial well-being [N= 1643] | .337 | .305 | .032 (.013) | 0.012 | ** | .025 (.012) | 0.034 | ** |
| Food insecurity [N= 1686] | .673 | .712 | -.039 (.024) | 0.095 | | -.015 (.022) | 0.486 | |
| Food satisfaction [N= 1683] | .287 | .237 | .050 (.016) | 0.002 | ** | .038 (.015) | 0.012 | ** |
| Poor physical or mental health [N= 1677] | .382 | .393 | -.011 (.020) | 0.594 | | -.002 (.017) | 0.921 | |
| Days absent from school | 12.3 | 13.0 | -.690 (.930) | 0.457 | | NA | NA | |
| Overall [N= 1602] | .112 | .000 | .112 (.038) | 0.003 | ** | .067 (.033) | 0.044 | ** |

Notes: ** P-value<.05. Results estimated using an OLS regression with robust standard errors. For financial well-being and food satisfaction, a positive difference reflects an improvement. For food insecurity, health, and days absent a negative value represents an improvement. The overall index is the mean of the first four measures after orienting them so that a positive difference represents an improvement and converting them into Z-scores.

**Table 8: Components of Financial Well-being Index (April 2021 Survey)**

| | | Without covariates | | | | With covariates | |
|---|---|---|---|---|---|---|---|
| | Index weight | Treatment | Control | Difference | P-value | Difference | P-value |
| Better off financially than six months ago | 0.333 | .268 | .208 | .060 (.022) | 0.008 ** | .060 (.022) | 0.005 ** |
| Did not have any bills, expenses, or needs that they were unable to pay | 0.167 | .430 | .424 | .007 (.026) | 0.798 | -.008 (.024) | 0.736 |
| Able to pay full utilities in March 2001 | 0.055 | .542 | .506 | .036 (.021) | 0.167 | .028 (.023) | 0.228 |
| Able to pay full rent in March 2001 | 0.055 | .685 | .698 | -.013 (.025) | 0.599 | -.018 (.022) | 0.408 |
| Did not delay health care or prescriptions because of costs | 0.055 | .862 | .860 | .002 (.019) | 0.919 | -.002 (.018) | 0.910 |
| Able to pay for $300 emergency expense with money I have | 0.333 | .183 | .154 | .030 (.019) | 0.118 | .018 (.019) | 0.341 |
| Total | 1.00 | .337 | .305 | .032 (.013) | 0.012 ** | .025 (.012) | 0.034 ** |

Notes: ** P-value<.05. N=1643.

**Table 9: Secondary Outcomes - Food Consumption**

| | | | Without covariates | | | With covariates | | |
|---|---|---|---|---|---|---|---|---|
| | Treatment | Control | Difference | | P-value | Difference | | P-value |
| Weekly food expenditures, $, (grocery, restaurant, delivery) [N= 1697] | 323.2 | 285.4 | 37.8 | (10.5) | 0.000 ** | 35.7 | (9.8) | 0.000 ** |
| Ate fresh vegetables *more* than once last week [N= 1701] | .625 | .581 | .044 | (.025) | 0.083 * | .043 | (.025) | 0.082 * |
| Ate fresh meat at least once last week [N= 1676] | .903 | .865 | .038 | (.017) | 0.031 ** | .038 | (.017) | 0.025 ** |
| Ate fish at least once last week [N= 1657] | .579 | .519 | .060 | (.026) | 0.022 ** | .057 | (.026) | 0.027 ** |
| Ate processed meat at least once last week [N= 1676] | .760 | .717 | .044 | (.023) | 0.061 * | .035 | (.023) | 0.127 |
| Ate legumes at least once last week [N= 1684] | .893 | .890 | .004 | (.016) | 0.827 | .005 | (.016) | 0.739 |
| Received free food boxes last week [N= 1686] | .723 | .673 | .050 | (.024) | 0.037 ** | .045 | (.023) | 0.056 * |
| Majority of household food consumption from free food boxes [N= 1180] | .459 | .537 | -.078 | (.031) | 0.012 ** | -.073 | (.031) | 0.020 ** |

Notes: ** P-value<.05 * P-value<.10.

**Table 10: Food Security - Experimental Impacts**

| | Without covariates | | | | With covariates | |
|---|---|---|---|---|---|---|
| | Treatment | Control | Difference | P-value | Difference | P-value |
| Household SOMETIMES or OFTEN NOT enough to eat, *December 2020* [N=1591] | .213 | .318 | -.105 (.024) | 0.000 ** | -.095 (.022) | 0.000 ** |
| Children SOMETIMES or OFTEN NOT enough to eat, *December 2020* [N=1221] | .476 | .527 | -.051 (.030) | 0.093 * | -.043 (.027) | 0.119 |
| Children SOMETIMES or OFTEN NOT enough to eat, *February 2021* [N=1101] | .415 | .487 | -.072 (.032) | 0.025 ** | -.046 (.028) | 0.099 * |
| Food Insecure from USDA 18-item test, *April 2021* [N=1686] | .673 | .712 | -.039 (.024) | 0.095 * | -.015 (.022) | 0.486 |
| USDA 18-item scaled score (0-10 range), *April 2021* [N=1686] | 3.52 | 3.63 | -.110 (.130) | 0.405 | .040 (.110) | 0.728 |
| Household SOMETIMES or OFTEN NOT enough to eat, *April 2021* [N=1701] | .170 | .190 | -.020 (.020) | 0.328 | -.009 (.019) | 0.619 |
| Children SOMETIMES or OFTEN NOT enough to eat, *April 2021* [N=1294] | .430 | .429 | .001 (.029) | 0.961 | .020 (.026) | 0.448 |

Notes: ** P-value<.05 * P-value<.10.

**Table 11: Components of Food Satisfaction Index (April 2021 Survey)**

| | | Without covariates | | | | With covariates | |
|---|---|---|---|---|---|---|---|
| | Index weight | Treatment | Control | Difference | P-value | Difference | P-value |
| Very satisfied with family's food situation | 0.333 | .249 | .218 | .031 (.022) | 0.157 | .009 (.021) | 0.662 |
| Family food situation is better than six months ago | 0.333 | .393 | .317 | .075 (.025) | 0.002 ** | .076 (.024) | 0.001 ** |
| Enough of the kinds of food we want | 0.333 | .220 | .177 | .043 (.020) | 0.034 ** | .028 (.020) | 0.158 |
| Total | 1.00 | .287 | .237 | .050 (.016) | 0.002 ** | .038 (.015) | 0.012 ** |

Notes: ** P-value<.05 * P-value<.10. N= 1683.

**Table 12: Components of Health Index (April 2021 Survey)**

| | | Without covariates | | | | With covariates | |
|---|---|---|---|---|---|---|---|
| | Index weight | Treatment | Control | Difference | P-value | Difference | P-value |
| Health fair or poor | 0.50 | .461 | .486 | -.025 (.026) | 0.328 | -.017 (.023) | 0.445 |
| Anxious or depressed | 0.50 | .304 | .299 | .004 (.024) | 0.862 | .014 (.022) | 0.528 |
| Total | 1.00 | .382 | .393 | -.011 (.020) | 0.594 | -.002 (.017) | 0.921 |

Notes: ** P-value<.05 * P-value<.10. N= 1677. "Anxious or depressed" = 1 if the respondent scored a 3 or above on either the PHQ-2 or GAD-2.

**Table 13: Experimental Impacts - Employment**

| | Without covariates | | | | With covariates | |
|---|---|---|---|---|---|---|
| | Treatment | Control | Difference | P-value | Difference | P-value |
| Respondent worked for pay last week | .446 | .422 | .024 (.026) | 0.354 | .002 (.021) | 0.916 |
| Respondent or spouse worked for pay last week | .552 | .528 | .023 (.021) | 0.363 | .002 (.022) | 0.914 |
| Respondent hours worked per week | 12.6 | 11.5 | 1.1 (.795) | 0.174 | .369 (.670) | 0.582 |
| Household hours worked per week | 18.7 | 17.5 | 1.2 (1.07) | 0.266 | .457 (.932) | 0.624 |

Notes: ** P-value<.05 * P-value<.10. N= 1693.

**Table 14: Experimental Impacts - Secondary Outcomes**

| | Without covariates | | | | | With covariates | | |
|---|---|---|---|---|---|---|---|---|
| | Treatment | Control | Difference | | P-value | Difference | | P-value |
| Paid for childcare [N= 1282] | .189 | .163 | .026 | (.022) | 0.236 | .013 | (.021) | 0.549 |
| Moved [N= 1677] | .084 | .124 | -.040 | (.016) | 0.014 ** | -.041 | (.016) | 0.011 ** |
| Pregnant [N= 1332] | .034 | .013 | .021 | (.008) | 0.009 ** | .019 | (.008) | 0.020 ** |
| Had COVID [N= 1682] | .431 | .389 | .042 | (.026) | 0.101 | .024 | (.023) | 0.287 |
| Someone in household had COVID [N= 1687] | .533 | .485 | .048 | (.026) | 0.063 * | .030 | (.023) | 0.184 |
| Spending on clothing [N= 1696] | 95.7 | 81.3 | 14.4 | (7.62) | 0.059 * | 12.6 | (7.27) | 0.084 * |
| Bank account balance [N= 1617] | 713.6 | 660.8 | 52.8 | (79.5) | 0.507 | 13.2 | (79.5) | 0.869 |
| Amount held in cash [N= 1653] | 82.7 | 48.1 | 34.6 | (19.1) | 0.070 * | 31.3 | (18.9) | 0.097 * |

Notes: ** P-value<.05 * P-value<.10.

**Table A1: Characteristics of Lottery Sample**

| | All (1) | | Lottery losers (2) | | Lottery winners (3) | | Lottery losers *weighted* (4) | | P-value on (3)-(4) difference (5) | Interview sample (6) | |
|---|---|---|---|---|---|---|---|---|---|---|---|
| Family size | 3.24 | (1.63) | 3.20 | (1.57) | 3.27 | (1.66) | 3.28 | (1.58) | .942 | 3.36 | (1.56) |
| Family income | $17,040 | (12,595) | $17,143 | (12,480) | $16,974 | (12,770) | $16,454 | (12,422) | .243 | $17,861 | (12,532) |
| Have at least one child ages 0 to 5 | .401 | (.491) | .377 | (.485) | .415 | (.493) | .423 | (.494) | .678 | 0.449 | (.498) |
| Have at least one child ages 6 to 17 | .564 | (.496) | .536 | (.499) | .582 | (.493) | .586 | (.493) | .849 | 0.603 | (.489) |
| Have at least one family member age 65+ | .194 | (.396) | .150 | (.358) | .222 | (.416) | .207 | (.405) | .316 | 0.152 | (.359) |
| Have at least one disabled family member | .193 | (.395) | .155 | (.363) | .217 | (.413) | .208 | (.406) | .560 | 0.163 | (.369) |
| Family not receiving UI | .800 | (.400) | .762 | (.426) | .824 | (.381) | .813 | (.390) | .376 | 0.794 | (.404) |
| Family not receiving other food assistance | .565 | (.496) | .526 | (500) | .591 | (.492) | .576 | (.494) | .398 | 0.524 | (.500) |
| No family member working | .560 | (.496) | .512 | (.500) | .591 | (.492) | .596 | (.491) | .775 | 0.504 | (.500) |
| At least one veteran in family | .018 | (.132) | .011 | (.106) | .022 | (.146) | .019 | (.137) | .674 | 0.007 | (.085) |
| Total tickets | 4.30 | (1.07) | 4.03 | (1.02) | 4.47 | (1.07) | 4.43 | (1.03) | .339 | 4.20 | (1.04) |
| N | 3615 | | 1402 | | 2213 | | 1402 | | | 1936 | |

Source: Authors' calculations from program application form data and baseline survey. Standard errors in parentheses. Column (5) P-values are estimated from regressions with robust standard errors and analytical weights. See text for explanation of weights.

**Table A.2: Survey Topics**

| Survey | Interview dates | Topics | | Completion Rate |
|---|---|---|---|---|
| Baseline survey | 9/11/2020 - 9/17/2020 | Demographics & household composition<br>Financial situation<br>Income and expenses | Food security<br>Health & healthcare<br>Employment | 100.0% |
| Pseudo baseline survey | 10/18/2020 - 11/20/2020 | Monthly financial situation<br>Employment status<br>Health and safety at work<br>Childcare<br>Food security and quality<br>Remittances | Receipt of benefits<br>Food expenditures<br>Food pantry use<br>Health & COVID<br>Internet access & online learning<br>Assets & debt | 92.0% |
| December pulse survey | 12/7/2020 - 12/20/2020 | Monthly financial situation<br>Employment status<br>Food security and free meals | | 89.4% |
| February pulse survey | 1/31/2021 - 2/16/2021 | Monthly financial situation<br>WTP for free meals | Mental health<br>Parenting & schooling | 89.4% |
| 24-hour food recall | 3/2021 - 5/2021 | ASA24 instrument | | N/A |
| Final survey | 4/12/2021 - 5/10/2021 | Monthly financial situation<br>Food security<br>Food pantry experience<br>Food quality<br>Food expenditures<br>Food satisfaction<br>Employment | Government benefits<br>Childcare<br>Online learning<br>Housing & moving<br>General health, mental health & COVID<br>Remittances<br>Assets & debt | 95.0% |

**Table A.3: Covariate Balance**

| | Lottery winners (1) | | Lottery losers (2) | | P-value on (1)-(2) difference (3) |
|---|---|---|---|---|---|
| **Age** | | | | | |
| <25 (omitted group) | | | | | |
| 25-29 | .108 | (.306) | .100 | (.293) | 0.606 |
| 30-34 | .156 | (.358) | .144 | (.343) | 0.489 |
| 35-39 | .177 | (.377) | .180 | (.376) | 0.897 |
| 40-44 | .153 | (.355) | .169 | (.367) | 0.377 |
| 45-49 | .110 | (.309) | .121 | (.319) | 0.488 |
| 50-54 | .085 | (.275) | .075 | (.257) | 0.441 |
| 55-59 | .057 | (.228) | .055 | (.224) | 0.894 |
| 60+ | .087 | (.278) | .097 | (.290) | 0.501 |
| **Female** | .803 | (.397) | .809 | (.393) | 0.785 |
| **Race and Ethnicity** | | | | | |
| Latino | .897 | (.299) | .895 | (.301) | 0.898 |
| White | .093 | (.284) | .088 | (.278) | 0.761 |
| Black | .028 | (.163) | .026 | (.155) | 0.739 |
| Other (omitted category) | | | | | |
| **Household Structure/Size** | | | | | |
| Number of adults in household | 2.39 | (1.23) | 2.46 | (1.21) | 0.269 |
| Number of kids under 5 | .506 | (.686) | .460 | (.651) | 0.170 |
| Number of kids 5 to 17 | 1.08 | (1.09) | 1.14 | (1.04) | 0.282 |
| **Financial situation vs. year ago** | | | | | |
| Better | .027 | (.157) | .025 | (.152) | 0.844 |
| Worse | .790 | (.395) | .796 | (.391) | 0.735 |
| The Same (omitted category) | | | | | |
| **Financial situation vs. last month** | | | | | |
| Better | .070 | (.245) | .067 | (.241) | 0.816 |
| Worse | .494 | (.482) | .481 | (.482) | 0.591 |
| The Same (omitted category) | | | | | |
| **Future financial situation - expectations in 6 mos.** | | | | | |
| Better | .279 | (.428) | .305 | (.440) | 0.224 |
| Worse | .208 | (.388) | .186 | (.371) | 0.252 |
| The Same (omitted category) | | | | | |
| **Financial hardship** - over past year, have you experienced any financial hardship? *("yes" response)* | .819 | (.372) | .808 | (.381) | 0.552 |
| **Sources of financial hardship** *(can have multiple)* | | | | | |
| Lost job | .454 | (.481) | .422 | (.477) | 0.179 |
| Spouse lost job | .182 | (.373) | .186 | (.375) | 0.863 |
| Reduced work hours | .274 | (.431) | .280 | (.434) | 0.783 |
| Spouse reduced work hours | .148 | (.343) | .144 | (.339) | 0.829 |
| Health emergency | .172 | (.364) | .158 | (.351) | 0.444 |
| **Other Financial Situations** | | | | | |
| Had unpaid bills in August 2020 | .688 | (.439) | .726 | (.415) | 0.067 |
| Currently behind on your rent or mortgage | .382 | (.457) | .371 | (.450) | 0.615 |
| Currently behind on your utility bills | .447 | (.463) | .439 | (.454) | 0.731 |
| Had any utilities disconnected in past 4 months | .414 | (.460) | .399 | (.455) | 0.528 |

**Table A.3: Covariate Balance**

| | Lottery winners (1) | | Lottery losers (2) | | P-value on (1)-(2) difference (3) |
|---|---|---|---|---|---|
| **Age** | | | | | |
| <25 (omitted group) | | | | | |
| 25-29 | .108 | (.306) | .100 | (.293) | 0.606 |
| 30-34 | .156 | (.358) | .144 | (.343) | 0.489 |
| 35-39 | .177 | (.377) | .180 | (.376) | 0.897 |
| 40-44 | .153 | (.355) | .169 | (.367) | 0.377 |
| 45-49 | .110 | (.309) | .121 | (.319) | 0.488 |
| 50-54 | .085 | (.275) | .075 | (.257) | 0.441 |
| 55-59 | .057 | (.228) | .055 | (.224) | 0.894 |
| 60+ | .087 | (.278) | .097 | (.290) | 0.501 |
| **Female** | .803 | (.397) | .809 | (.393) | 0.785 |
| **Race and Ethnicity** | | | | | |
| Latino | .897 | (.299) | .895 | (.301) | 0.898 |
| White | .093 | (.284) | .088 | (.278) | 0.761 |
| Black | .028 | (.163) | .026 | (.155) | 0.739 |
| Other (omitted category) | | | | | |
| **Household Structure/Size** | | | | | |
| Number of adults in household | 2.39 | (1.23) | 2.46 | (1.21) | 0.269 |
| Number of kids under 5 | .506 | (.686) | .460 | (.651) | 0.170 |
| Number of kids 5 to 17 | 1.08 | (1.09) | 1.14 | (1.04) | 0.282 |
| **Financial situation vs. year ago** | | | | | |
| Better | .027 | (.157) | .025 | (.152) | 0.844 |
| Worse | .790 | (.395) | .796 | (.391) | 0.735 |
| The Same (omitted category) | | | | | |
| **Financial situation vs. last month** | | | | | |
| Better | .070 | (.245) | .067 | (.241) | 0.816 |
| Worse | .494 | (.482) | .481 | (.482) | 0.591 |
| The Same (omitted category) | | | | | |
| **Future financial situation - expectations in 6 mos.** | | | | | |
| Better | .279 | (.428) | .305 | (.440) | 0.224 |
| Worse | .208 | (.388) | .186 | (.371) | 0.252 |
| The Same (omitted category) | | | | | |
| **Financial hardship** - over past year, have you experienced any financial hardship? *("yes" response)* | .819 | (.372) | .808 | (.381) | 0.552 |
| **Sources of financial hardship** *(can have multiple)* | | | | | |
| Lost job | .454 | (.481) | .422 | (.477) | 0.179 |
| Spouse lost job | .182 | (.373) | .186 | (.375) | 0.863 |
| Reduced work hours | .274 | (.431) | .280 | (.434) | 0.783 |
| Spouse reduced work hours | .148 | (.343) | .144 | (.339) | 0.829 |
| Health emergency | .172 | (.364) | .158 | (.351) | 0.444 |
| **Other Financial Situations** | | | | | |
| Had unpaid bills in August 2020 | .688 | (.439) | .726 | (.415) | 0.067 |
| Currently behind on your rent or mortgage | .382 | (.457) | .371 | (.450) | 0.615 |
| Currently behind on your utility bills | .447 | (.463) | .439 | (.454) | 0.731 |
| Had any utilities disconnected in past 4 months | .414 | (.460) | .399 | (.455) | 0.528 |

**Table A.3: Covariate Balance, continued**

| | Lottery winners (1) | | Lottery losers (2) | | P-value on (1)-(2) difference (3) |
|---|---|---|---|---|---|
| **USDA Food Insecurity Index** | 3.46 | (1.89) | 3.60 | (1.86) | 0.142 |
| **The following describes food eaten in your house over the last 4 mos.** | | | | | |
| Enough of the kinds of food we want (omitted category) | | | | | |
| Enough but not always the kinds of food we want | .384 | (.455) | .372 | (.448) | 0.588 |
| Sometimes not enough to eat | .388 | (.456) | .408 | (.456) | 0.375 |
| Often not enough to eat | .103 | (.283) | .113 | (.295) | 0.478 |
| **Children not eating enough because couldn't afford food** | | | | | |
| Often true | .104 | (.283) | .109 | (.288) | 0.737 |
| Sometimes true | .328 | (.436) | .336 | (.435) | 0.712 |
| Never true (omitted category) | | | | | |
| **Monthly income before Covid adjusted for household size** | $ 1,312 | (634) | $ 1,214 | (662) | 0.003 |
| **Ratio of current monthly income to pre-Covid income** | .706 | (.349) | .713 | (.351) | 0.715 |
| **Health** | | | | | |
| Excellent | .086 | (.262) | .069 | (.232) | 0.150 |
| Very good | .166 | (.344) | .200 | (.373) | 0.067 |
| Good (omitted category) | | | | | |
| Fair | .342 | (.440) | .342 | (.439) | 0.977 |
| Poor | .072 | (.240) | .073 | (.241) | 0.924 |
| **Kessler 6 Mental Health Scale** | 10.5 | (5.24) | 10.9 | 5.48 | 0.104 |
| **Has had COVID** | .324 | (.427) | .315 | (.423) | 0.684 |
| **Unemployment** | | | | | |
| Unemployed looking for work | .444 | (.449) | .452 | (.448) | 0.749 |
| Unemployed not looking for work | .189 | (.355) | .197 | (.360) | 0.674 |
| Not working disabled | .145 | (.321) | .141 | (.317) | 0.819 |
| Not working home responsibilities | .180 | (.348) | .183 | (.350) | 0.840 |
| **N** | **1067** | | **730** | | |

Source: Authors' calculations from baseline survey. Standard deviations in parentheses. Column (3) P-values are estimated from regressions with robust standard errors and analytical weights. See text for explanation of weights. In addition to variables shown, regression covariates include squared values of adjusted income and the income ratio. Monthly income adjusted for household size is calculated using the U.S. Census Bureau three-parameter equivalence scale.

| Table A.4: Job Categories | | | |
|---|---|---|---|
| (% of respondents in each job) | | | |
| **Food Industry** | **30.1%** | **Education/Childcare** | **6.4%** |
| Cook | 18.4% | Daycare teacher/staff | 2.1% |
| Food packaging | 6.4% | Nanny | 2.0% |
| Restaurant | 1.8% | Teacher | 0.5% |
| Dishwasher | 1.7% | Teacher's aide | 0.2% |
| Food prep | 1.7% | Preschool teacher | 0.2% |
| **Housekeeping/Janitorial** | **26.2%** | Staff at library | 0.1% |
| Cleaning | 20.8% | High school teacher | 0.1% |
| Housekeeping | 2.9% | Education | 0.1% |
| Maintenance | 1.0% | School aide | 0.1% |
| House cleaner | 1.0% | Family and school liaison | 0.1% |
| Concierge | 0.5% | Education paraprofessional | 0.1% |
| **Customer service** | **10.6%** | Library aide | 0.1% |
| Cashier | 7.4% | After school teacher | 0.1% |
| Customer service | 1.1% | Youth worker | 0.1% |
| Sales | 0.9% | Assistant librarian | 0.1% |
| Laundry | 0.6% | Babysitter | 0.1% |
| Hair stylist | 0.6% | **Transportation** | **3.9%** |
| **Construction** | **7.5%** | Driver | 2.5% |
| Construction | 3.7% | Bus monitor | 0.5% |
| Laborer | 1.4% | Uber driver | 0.4% |
| Carpenter | 1.1% | Bus driver | 0.4% |
| Painter | 0.9% | Truck driver | 0.2% |
| Landscaper | 0.5% | **Warehouse/Factory** | **3.3%** |
| **Healthcare** | **4.4%** | Factory worker | 1.7% |
| Nurse assistant | 1.6% | Warehouse | 0.5% |
| PCA | 1.0% | Machine operator | 0.4% |
| Home health care | 0.7% | Quality Control | 0.5% |
| Home health aide | 0.6% | Warehouse worker | 0.2% |
| Nurse | 0.5% | **Other** | **4.5%** |
| **Clerical** | **3.1%** | Other | 1.7% |
| Secretary | 1.2% | Retail | 0.6% |
| Administrative Assistant | 0.6% | Homemaker | 0.6% |
| Administration | 0.6% | Security guard | 0.6% |
| Clerk | 0.4% | Production | 0.5% |
| Office clerk | 0.2% | Supervisor | 0.5% |

**Table A.5: Government Benefits**

| Baseline survey - September 2020 | |
|---|---|
| **What kind of health insurance or health care coverage do you have? [N= 1670]** | |
| None | 11.0% |
| Private insurance through my employer | 6.3% |
| Private insurance through spouse's employer | 2.6% |
| Massachusetts Health Connector *(ACA health insurance exchange)* | 22.7% |
| Medicaid/MassHealth | 48.3% |
| Military or veteran's plan | 0.2% |
| State-sponsored plan or other government health plan | 5.5% |
| Health insurance through college/university | 0.1% |
| Other | 3.3% |
| **November 2020 Survey** | |
| **Last month, did anyone in this household receive benefits from SNAP? [N=1762]** | |
| Yes | 46.6% |
| No | 53.4% |
| **Who received SNAP?** | |
| Both adults and children | 57.6% |
| Adults only | 4.9% |
| Children only | 37.5% |
| **Last month, did you or anyone in your household receive any of the following other government benefits? [N= 1720]** | |
| WIC | 20.8% |
| Unemployment insurance | 11.7% |
| Housing assistance | 7.3% |
| TANF | 1.0% |
| MassHealth (Medicaid) | 21.7% |
| MassHealth Limited | 25.9% |
| MassHealth CHIP | 10.6% |
| Medicare | 5.9% |
| Social Security Retirement | 3.3% |
| Social Security Disability | 5.1% |
| SSI | 4.6% |
| P-EBT | 28.3% |
| None of these | 21.8% |

Source: Authors' calculations from September and November 2020 surveys.

**Table A.6: Remittances, Gifts Received, Free Food**

| November 2020 Survey | | |
|---|---|---|
| **Thinking back to the time before Covid, how often did anyone in this household give or send money to relatives or friends living OUTSIDE the U.S.? [N= 1767]** | | |
| Every month | | 30.7% |
| Most months | | 9.2% |
| About half of months | | 3.1% |
| A few times a year | | 19.0% |
| Once a year | | 5.2% |
| Never | | 32.9% |
| **Mean amount sent per month [N= 1106]** | **$** | **279** |
| **Last month, did anyone in this household give or send money to relative or friends living outside the U.S.? [N= 1171]** | | |
| Yes | | 26.8% |
| No | | 73.2% |
| **Mean amount sent last month [N= 290]** | **$** | **215** |
| **Last month, did you or anyone in your household give or send money to relatives or friends living inside the U.S.? [N= 1752]** | | |
| Yes, we gave money to somone in Chelsea | | 3.5% |
| Yes, we gave money to someone in Massachusetts, but not in Chelsea | | 2.5% |
| Yes, we send money to someone inside the U.S., but not in Massachusetts | | 3.0% |
| No, we did not send money to others in the U.S. | | 91.8% |
| **Mean amount sent last month [N= 136]** | **$** | **223** |
| **Last month, did anyone in this household receive money from relatives or friends? [N= 1753]** | | |
| Yes | | 8.7% |
| No | | 91.3% |
| **Mean amount received last month [N= 147]** | | **$331** |
| **During the last 7 days, did you or anyone else in your household get free groceries or a free meal? [N= 1747]** | | |
| Yes | | 63.9% |
| No | | 36.1% |
| **(For those who got free food) How much of the food your household ate last week came from free meals? [N= 1099]** | | |
| All | | 6.4% |
| Most | | 27.2% |
| Half | | 21.6% |
| Some | | 31.1% |
| Very little | | 13.7% |
| **Where did you get free groceries or free meals? Select all that apply. [N= 1111]** | | |
| Free meals through the school or other programs aimed at children | | 33.1% |
| Food pantry or food bank | | 42.5% |
| Home-delivered meal service like Meals on Wheels | | 1.7% |
| Church, synagogue, temple, mosque or other religious organization | | 23.7% |
| Shelter or soup kitchen | | 2.8% |
| Other community program | | 44.6% |
| Family, friends, or neighbors | | 8.1% |

Source: Authors' calculations from November 2020 survey.

**Table A.7: Assets & Debt**

| November 2020 Survey | |
|---|---|
| **Do you or any members of your household have a bank account? [N= 1731]** | |
| Yes | 56.4% |
| No | 43.6% |
| Current total bank account value (mean) [N= 895] | $ 1,132 |
| **Do you or any members of your household keep any savings in cash? [N= 1706]** | |
| Yes | 10.4% |
| No | 89.6% |
| **Current total savings in cash (mean) [N=172]** | **$ 795** |
| **Do you or any members of your household have any retirement accounts such as 401(k)s, 403(b)s, or IRAs? [N= 1710]** | |
| Yes | 5.6% |
| No | 94.4% |
| **Current total value of all retirement accounts (mean) [N= 83]** | **$ 10,182** |
| **Do you or anyone else in your household have any other financial assets such as mutual funds, directly-held stocks, or bonds? [N= 1694]** | |
| Yes | 0.4% |
| No | 99.6% |
| **Current total value of all other financial assets [N= 6]** | **$ 2,208** |
| **Do you or anyone else in your household have any credit cards including store cards and gas cards? [N= 1722]** | |
| Yes | 30.4% |
| No | 69.6% |
| **Current total credit card debt (mean) [N= 404]** | **$ 5,025** |
| **Do you or anyone in your household have any student loans? [N= 1734]** | |
| Yes | 7.2% |
| No | 92.8% |
| **Current student loan debt (mean) [N= 124]** | **$ 17,629** |
| **Do you or anyone in your household have any debt from medical bills? [N= 1725]** | |
| Yes | 18.0% |
| No | 82.0% |
| **Current total medical debt (mean) [N= 310]** | **$ 2,185** |
| **Do you or anyone in your household have any other debt such as personal loans from friends, family members, or anyone else? [N= 1735]** | |
| Yes | 24.8% |
| No | 75.2% |
| **Current debt from personal loans (mean) [N= 426]** | **$ 4,259** |

Source: Authors' calculations from November 2020 survey.

**Table A.8: Secondary Outcomes - Financial Well-Being**

| | | | Without covariates | | | With covariates | | |
|---|---|---|---|---|---|---|---|---|
| **April 2021** | Treatment | Control | Difference | | P-value | Difference | | P-value |
| Financial situation BETTER than last month | .223 | .177 | .045 | (.021) | 0.027 ** | .042 | (.020) | 0.037 ** |
| Financial situation future outlook better | .429 | .395 | .035 | (.025) | 0.171 | .038 | (.024) | 0.115 |
| Phone or utilities disconnected | .461 | .436 | .025 | (.026) | 0.321 | .028 | (.024) | 0.250 |
| Past due rent | .345 | .326 | .019 | (.025) | 0.442 | .019 | (.022) | 0.384 |
| Past due bills/utilities | .459 | .478 | -.020 | (.026) | 0.447 | -.018 | (.024) | 0.436 |
| Confident can pay rent next month | .355 | .367 | -.012 | (.025) | 0.638 | -.017 | (.023) | 0.456 |
| Eviction concern | .192 | .198 | -.006 | (.021) | 0.770 | -.006 | (.020) | 0.777 |
| **February 2021** | | | | | | | | |
| Financial situation BETTER than last month | .187 | .097 | .090 | (.018) | 0.000 ** | .099 | (.018) | 0.000 ** |
| Past due rent | .633 | .596 | .038 | (.026) | 0.149 | .036 | (.024) | 0.134 |
| Confident can pay rent next month | .303 | .308 | -.005 | (.025) | 0.853 | -.007 | (.023) | 0.758 |
| Eviction concern | .217 | .245 | -.028 | (.023) | 0.217 | -.032 | (.022) | 0.140 |
| **December 2020** | | | | | | | | |
| Financial situation BETTER than last month | .123 | .055 | .068 | (.014) | 0.000 ** | .069 | (.014) | 0.000 ** |
| Past due rent | .636 | .633 | .003 | (.026) | 0.919 | .000 | (.023) | 0.984 |
| Confident can pay rent next month | .261 | .250 | .011 | (.023) | 0.625 | .007 | (.022) | 0.742 |
| Eviction concern | .217 | .235 | -.019 | (.023) | 0.414 | -.020 | (.022) | 0.355 |

Notes: ** P-value<.05 * P-value<.10